

'In-Between' Uncertainty in Bayesian Neural Networks

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Task/Model Motivation

- ▶ Bayesian Neural networks underestimate uncertainty
- ▶ Reliable uncertainty estimates are critical for:
 - ▶ Out-of-distribution robustness
 - ▶ Critical decision-making tasks
- ▶ Networks should predict with high uncertainty:
 - ▶ Far from training data
 - ▶ *Between separated clusters of observations*
- ▶ Mean-Field Variational Inference (MFVI) is widely used but has limitations

The In-Between Uncertainty Problem

What does “in-between uncertainty” mean?

- ▶ Consider two separated clusters of observed data
- ▶ A well-calibrated model should:
 - ▶ Have **high confidence** near data clusters
 - ▶ Have **low confidence (high uncertainty)** in the gap between clusters
- ▶ Standard UCI benchmarks don't test this property
- ▶ MFVI fails catastrophically on this requirement

The In-Between Uncertainty Problem

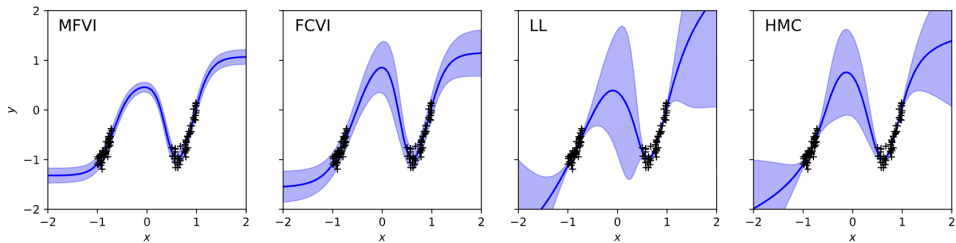


Figure 1. Mean and two standard deviation bars of the predictive distribution for $f_\theta(x)$ (without output noise).

Formal Problem Statement: Setup

- ▶ Bayesian Neural Network with parameters θ
- ▶ Output $f_{\theta}(x)$ interpreted as mean of Gaussian with noise variance σ_o^2
- ▶ Dataset: $\mathcal{D} = (x_n, y_n)_{n=1}^N$
- ▶ Prior: $p(\theta)$ (diagonal Gaussian)
- ▶ Posterior: $p(\theta|\mathcal{D})$
- ▶ Goal: Approximate posterior for predictions

Problem: How does approximate inference affect uncertainty quality *between* data clusters?

Predictive Distribution

The predictive distribution is:

$$p(y^*|x^*, \mathcal{D}) = \mathbb{E}_{p(\theta|\mathcal{D})}[p(y^*|x^*, \theta)]$$

With approximate posterior $q_\phi(\theta)$:

$$p(y^*|x^*, \mathcal{D}) \approx \mathbb{E}_{q_\phi(\theta)}[p(y^*|x^*, \theta)] \approx \frac{1}{M} \sum_{m=1}^M p(y^*|x^*, \theta_m)$$

Question: Does the choice of $q_\phi(\theta)$ affect uncertainty *between* data regions?

Mean-Field Variational Inference (MFVI)

Approach: Approximate $p(\theta|\mathcal{D})$ with $q_\phi(\theta)$

Optimize Evidence Lower Bound (ELBO):

$$\mathcal{L}(\phi) = \sum_{n=1}^N \mathbb{E}_{q_\phi} [\log p(y_n|\theta, x_n)] - \text{KL}(q_\phi(\theta) || p(\theta))$$

$$\mathcal{L}(\phi) \approx \sum_{m=1}^M \sum_{n=1}^N \mathbb{E}_{q_\phi} [\log p(y_n|\theta_m, x_n)] - \text{KL}(q_\phi(\theta) || p(\theta))$$

Mean-Field Assumption: $q_\phi(\theta)$ is fully factorized Gaussian

$$q_\phi(\theta) = \prod_i \mathcal{N}(\theta_i; \mu_i, \sigma_i^2)$$

Problem: Independence assumption restricts expressiveness

Laplace Approximation

Classical alternative: Gaussian approximation at MAP solution

1. Find mode: $\theta_{\text{MAP}} = \arg \max_{\theta} p(\theta|\mathcal{D})$
2. Set $q(\theta) = \mathcal{N}(\theta; \theta_{\text{MAP}}, \Sigma)$
3. Covariance from curvature:

$$\Sigma = - [\nabla_{\theta} \nabla_{\theta} \log p(\theta|\mathcal{D})|_{\theta=\theta_{\text{MAP}}}]^{-1}$$

In practice: Use Gauss-Newton matrix

$$\Sigma = - \left[\frac{1}{\sigma_o^2} \sum_{n=1}^N g(x_n) g(x_n)^T + \text{diag}(p) \right]^{-1}$$

where $g(x_n) = \nabla_{\theta} f_{\theta}(x_n)|_{\theta=\theta_{\text{MAP}}}$ and $p_i = 1/\sigma_i^2$ and σ_i^2 is prior variance of θ_i

Linearized Laplace (LL) Approximation

Key improvement: Linearize network output around θ_{MAP}

$$f_{\theta}(x) \approx f_{\theta_{\text{MAP}}}(x) + g(x)^T(\theta - \theta_{\text{MAP}})$$

This gives a *linear-Gaussian model* with exact solution:

$$p(y_*|x_*, \mathcal{D}) \approx \mathcal{N}(y_*; f_{\theta_{\text{MAP}}}(x_*), \sigma_o^2 + g(x_*)^T \Sigma g(x_*))$$

Advantages:

- ▶ Full covariance structure
- ▶ Similar underfitting of standard Laplace
- ▶ Still tractable for small networks

Synthetic 1D Experiment

Setup:

- ▶ Sine wave with Gaussian noise
- ▶ Two separated clusters of observations
- ▶ 1 hidden layer, 50 units, tanh activation

Methods compared:

- ▶ MFVI (mean-field)
- ▶ FCVI (full covariance VI)
- ▶ LL (Linearized Laplace)
- ▶ HMC (Hamiltonian Monte Carlo - ground truth)

Synthetic 1D Experiment: Results

Key observations from Figure 1 in paper:

- ▶ **MFVI**: Error bars similar magnitude in data region and in-between region
 - ▶ Fails to capture in-between uncertainty
- ▶ **FCVI**: Larger uncertainty in middle, slightly underconfident in data region
- ▶ **LL**: High confidence in data, increased uncertainty between
 - ▶ Matches HMC behavior
- ▶ **HMC**: Ground truth

UCI Regression Datasets

Two experimental setups:

1. **Standard splits:** Uniform random train/test split
 - ▶ Tests interpolation and general performance
2. **Gap splits:** Remove middle 1/3 of data for testing
 - ▶ Tests in-between uncertainty explicitly
 - ▶ For each input dimension, sort and create gap

Evaluation metric: Test log-likelihood

- ▶ Penalizes both inaccuracy and overconfidence

UCI Results: Standard Splits

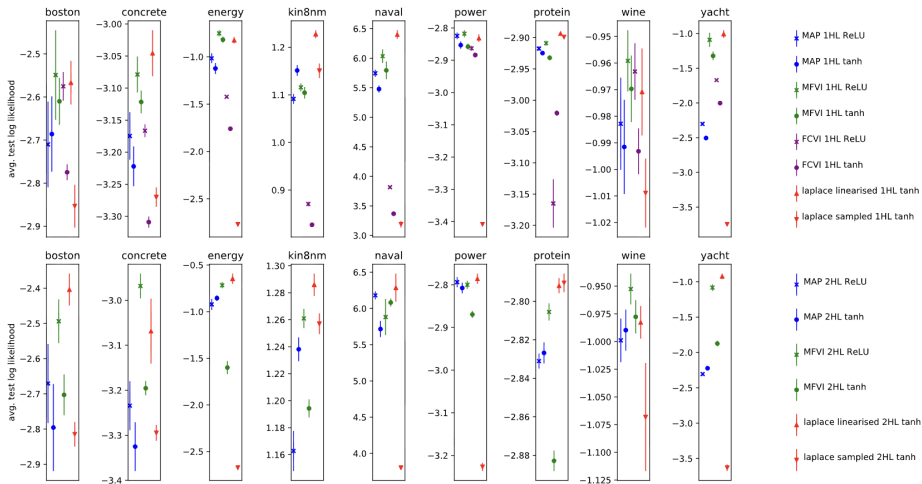


Figure 2. Average test log-likelihoods on the standard splits for BNNs with one hidden layer (top) and two hidden layers (bottom). There are 50 hidden units in each layer.

UCI Results: Standard Splits

Performance on standard splits (Figure 2):

- ▶ MFVI and LL perform best overall
- ▶ MAP competitive on most datasets
- ▶ FCVI struggles (optimization difficulties)
- ▶ Both 1 and 2 hidden layer networks tested

Conclusion: MFVI appears to work well for standard benchmarks

UCI Results: Standard Splits

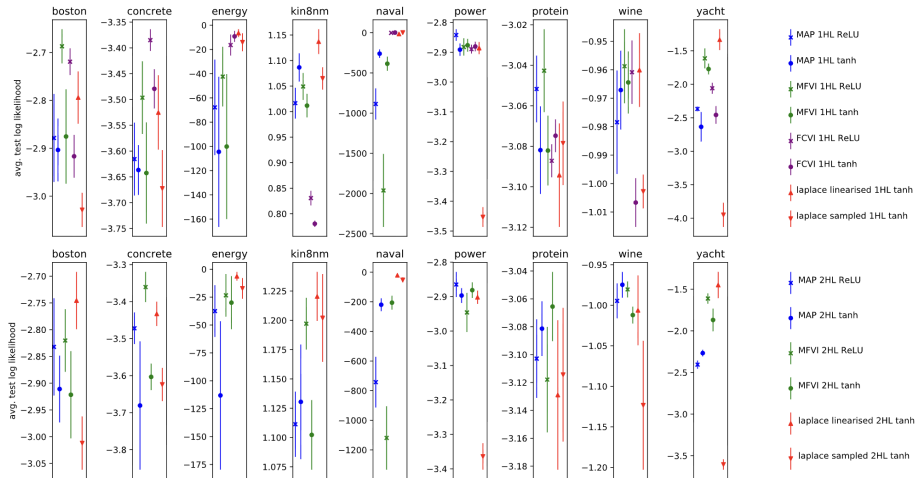


Figure 3. Average test log-likelihoods on the gap splits for BNNs with one hidden layer (top) and two hidden layers (bottom). Note the scale on energy and naval, where MAP and MFVI fail catastrophically. There are 50 hidden units in each layer.

UCI Results: Gap Splits - Catastrophic Failures

Performance on gap splits (Figure 3):

- ▶ Most datasets: MAP competitive with Bayesian methods
- ▶ **Energy dataset:**
 - ▶ MAP: catastrophic failure
 - ▶ MFVI: catastrophic failure
 - ▶ LL: close to ground truth
- ▶ **Naval dataset:**
 - ▶ MAP: catastrophic failure
 - ▶ MFVI: catastrophic failure
 - ▶ LL: close to ground truth

Key insight: MFVI is overconfident, performs similarly to MAP

Literature

Key references:

- ▶ MacKay (1992): Laplace approximation for BNNs
- ▶ Blundell et al. (2015): Bayes by Backprop with reparameterization
- ▶ Gal Ghahramani (2016): Monte Carlo Dropout
- ▶ Hernández-Lobato Adams (2015): Probabilistic Backpropagation
- ▶ Ritter et al. (2018): Scalable Laplace with K-FAC
- ▶ Lawrence (2001): Underfitting problem in standard Laplace

Paper: Foong et al., ICML 2019 Workshop on Uncertainty and Robustness